

JOEL SOMANJE

COMPUTER NETWORK ENGINEERING STUDENT

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PROFILE

Motivated fourth-year Bachelor of Science in Computer Network Engineering student at the University of Malawi, expected to graduate in August 2027. I have practical experience in computer networking, routing and switching, network security, troubleshooting, Linux administration, programming, containerization and artificial intelligence projects.

I have used Cisco Packet Tracer, GNS3, Wireshark, PuTTY, Python, Java, C++, Bash/Shell scripting, Git, Docker, YAML and Ollama, and I am seeking an industrial attachment or internship opportunity in telecommunications, network support, ICT infrastructure, IT support, network security, network engineering, cybersecurity or artificial intelligence.

EDUCATION

University of Malawi

Bachelor of Science in Computer Network Engineering — Fourth Year

Expected Graduation: August 2027

TECHNICAL SKILLS

- Routing and Switching, Network Security, Network Troubleshooting – design, connectivity, traffic analysis and configuration troubleshooting in simulated environments
- Cisco Packet Tracer / GNS3 – Network simulation, design, configuration and testing; Wireshark – packet capture and protocol analysis
- PuTTY – Remote access and CLI administration; Linux Systems Administration – command line administration and Bash/Shell scripting; Docker – containerization
- Python, Java, C++ – Programming, AI projects and Retrieval-Augmented Generation systems; Git/GitHub – version control; YAML – configuration files
- Artificial Intelligence – Machine-learning fundamentals, RAG systems and local Large Language Model applications using Ollama
- Network Programming – Client-server socket application development using Windows Sockets (Winsock)

SOFT SKILLS

Infrastructure management, collaborative problem-solving, technical mentorship, teamwork and communication, technical documentation, and research and analytical thinking.

PROJECTS

Enterprise Airport Network — github.com/JoelFrankSomanje/enterprise-airport-project. Enterprise network design project demonstrating practical routing, switching, network architecture, connectivity and troubleshooting concepts in a simulated airport environment.

AI-Powered Cisco Configuration Auditor — Networking and AI project focused on applying Python and artificial intelligence concepts to analyse and audit Cisco device configurations.

Retrieval-Augmented Generation System — Python and Ollama project involving the development of a RAG system for working with contextual information and locally run Large Language Models.

Networking Projects and Practical Labs — github.com/JoelFrankSomanje/Networking-Projects-No-fuss.

Practical networking work involving routing and switching labs using Cisco Packet Tracer, network design and troubleshooting using GNS3, and network-security and traffic-analysis labs using Wireshark.

Client-Server Application — Java/C++ and Windows Sockets (Winsock) project involving the development and testing of a client-server network application.

Programming and Linux Administration Labs — github.com/JoelFrankSomanje/Cplusplus-Projects. Practical Java and C++ programming projects, Linux administration exercises, Bash/Shell scripting, Git-based project work and systems troubleshooting activities.

CERTIFICATIONS

Introduction to Java – SoloLearn; Getting Started with Cisco Packet Tracer – Cisco Networking Academy; Linux Administration Workshop – Hack The Box.

REFEREES

Mr. Samuel Yute — Lecturer, University of Malawi — Tel: +265 881 481 182

Mr. Carlos Muleke — COMSOC (Computer Society) Project Coordinator — Tel: +265 992 779 377

RELEVANT COURSEWORK

First Year:

Introduction to Computer Science, Introduction to Computer Programming, College Algebra, Trigonometry and Elementary Calculus, Descriptive Statistics and The Statistical System.

Second Year:

Introduction to Computer Network Engineering, Advanced Computer Programming, Database Systems, Operating Systems, Discrete Mathematics, Analogue Electronics, Digital Electronics 1, Device Electronics, Digital Electronics 2 and Network Analysis.

Third and Fourth Year:

Algorithms and Data Structures, Introduction to Cryptography, Linux Systems Administration, Automata Theory, Computer Networks, Artificial Intelligence, Network Routing and Switching, Network Programming and Application Development, Project Management, Research Methods and Ethics in Computing, Network Security, Network Monitoring and Bandwidth Optimisation, and Wireless and Mobile Networks.